# Can an LLM Judge Replace Human Coders for Measuring Scientific-Evidence Use in Policy? A Feasibility Study and a Cautionary Result

Nitheesha Nakka Bruce A. Desmarais Sarah Rajtmajer Jeffrey J. Harden

Frederick J. Boehmke

June 26, 2026

**Abstract**

Can we measure, at scale and *without human coders*, whether the scientific evidence that legislators invoke supports the claims they make? We build an end-to-end pipeline that (i) extracts empirical claims from legislators' press releases, (ii) retrieves candidate peer-reviewed science, and (iii) assigns each (claim, reference) pair a stance (support/refute/silent). The hard step is the third: assessing *support* has historically required scarce expert coders. We test a fully autonomous alternative—a high-capacity large language model (LLM) as an in-context judge providing "gold" labels, independent open-source natural-language-inference (NLI) classifiers as a scalable surrogate, and prediction-powered inference (PPI) / design-based supervised learning (DSL) to combine them. We report this as a *feasibility study*, not a validated measurement, because we deliberately use *no human-coded labels*: the foundations we build on (the alternative-annotator test; DSL/PPI) formally require a trusted human anchor, which we lack, so our estimates are conditional on a single, unvalidated LLM judge. Two results follow, and we read them cautiously. First, our *zero-shot* NLI configuration agrees with the LLM judge only weakly ($\kappa \approx 0.15$–$0.21$) and labels most pairs "silent"; anchoring on the LLM judge and applying PPI moves the estimated support rate from a naive 0.21/0.11 (D/R) to 0.71/0.54 "under the LLM judge." We argue this $\sim 3\times$ gap is too large to attribute cleanly to NLI error: it is equally consistent with a misconfigured NLI baseline and with the LLM judge *over*-calling support—and, because the claims and references were themselves produced by LLMs and our judge is an LLM, with *self-preference* (a documented bias) inflating the result. Self-preference and the absence of a human anchor are first-order threats, not footnotes. Our contribution is therefore a reusable, no-API, no-RA *template* for this kind of measurement, together with a cautionary result about what it can and cannot establish on its own.

## 1 Introduction

Legislators routinely reach for scientific authority. A bill is "backed by research," a problem is established with a statistic, an intervention is said to be "proven" to work. In technically complex domains—and few are more complex or fast-moving than artificial intelligence (AI)—these appeals to evidence are both more common and harder to evaluate. Yet a basic question has remained effectively unmeasurable at scale: when a legislator invokes science to support a claim, does the scientific record actually support that claim? Answering it would let us move the study of research use in policy (Weiss, 1979; Oliver et al., 2022) from case studies and surveys toward systematic, population-level measurement.

The obstacle is not data but *validation*. Legislators' public communications are abundant and machine-readable, and modern language models can extract the empirical claims they contain and retrieve plausibly relevant scientific literature. The hard step is the last one: deciding whether a retrieved study *supports*, *refutes*, or is *silent* on a given claim. This judgment has traditionally required trained human coders, whose cost and scarcity cap the size and scope of any such study and whose reliability must itself be established. As a practical matter, the "does-the-science-support-the-claim" step is where projects of this kind stall.

We show that this step can be performed *autonomously*, with no human research assistants and no proprietary application programming interfaces (APIs), while preserving statistical validity. Our design rests on three ideas from the recent literature. First, high-capacity large language models (LLMs) now match or exceed trained crowd coders on relevance and stance tasks (Gilardi et al., 2023; Ziems et al., 2024), so an LLM can supply gold-standard labels. Second, because any machine labels are imperfect and naively substituting them for human labels biases downstream estimates (Egami et al., 2023), we combine a small set of high-quality LLM-judge labels with a large set of cheap, independent classifier labels using prediction-powered inference (PPI) (Angelopoulos et al., 2023) and design-based supervised learning (DSL) (Egami et al., 2023). Third, because any single judge can be biased (Zheng et al., 2023), we pair a generative LLM judge with discriminative natural-language-inference (NLI) classifiers that fail in different ways, and we report their agreement explicitly, in the spirit of the alternative-annotator test (Calderon et al., 2025).

We apply the system to a corpus of 7,139 California State Assembly press releases, 303 of which discuss AI, and to a parallel pilot in K–12 education policy, and we report two results cautiously. *First, methodologically,* our *zero-shot* NLI configuration agrees with the LLM judge only weakly ($\kappa \approx 0.15$–$0.21$) and labels most pairs "silent," because strict entailment between an abstract and a press-release claim is usually "neutral" even when the study substantiates the claim's substance. Anchoring on the LLM judge and applying PPI moves the estimated support rate from a naive 0.21 (Democrats) / 0.11 (Republicans) to 0.71 / 0.54. We resist the tempting reading that "the LLM judge recovers the truth." A correction this large ($\sim 3\times$) is atypical when a surrogate is competent, and is equally consistent with a misconfigured NLI baseline (zero-shot entailment is a poor proxy for scientific support; serious claim verification uses fine-tuning and rationale selection (Wadden et al., 2020)) and with the LLM judge *over*-calling support. Because the claims and references were produced by LLMs and our judge is an LLM, *self-preference*—the documented tendency of LLM evaluators to favor model-generated text (Panickssery et al., 2024)—could inflate the support rate, and we have no human anchor to rule this out. *Second, substantively and provisionally,* under our judge both parties' AI claims are supported at broadly similar (and fairly high) rates, so the large partisan gap implied by the naive NLI estimate does not survive; what is robust is that Democrats discuss AI far more often and more intensively than Republicans, not any difference in the support for their claims.

Our contributions are deliberately modest. (1) We provide a reusable, end-to-end, fully autonomous *template*—claim extraction, evidence retrieval, and support assessment—that runs with no human coders and no paid APIs, released as open-source software, that others can apply and, crucially, validate against a human standard. (2) We show that a popular shortcut—scoring scientific support with off-the-shelf zero-shot NLI—can be badly miscalibrated for press-release claims, a cautionary result for the growing practice of measuring political text with such tools. (3) We are explicit about what an autonomous, single-model, self-judging pipeline *cannot* establish on its own (validity against truth), and we specify what a valid version would require. We do *not* claim a validated estimate of evidence use; the substantive patterns we report are provisional and conditional on an unvalidated judge. The remainder of the paper describes the related literatures (§2), the data (§3), our methods (§4), and our results (§5), before discussing implications and limitations (§6–§7).

## 2 Related work

**Evidence use in policymaking.** A long line of scholarship asks how, and how much, research evidence informs policy. Weiss (1979) distinguished instrumental, conceptual, and symbolic uses of research; more recent work emphasizes the organizational and relational conditions under which evidence is taken up (Oliver et al., 2022; Bogenschneider et al., 2019). This literature is overwhelmingly qualitative or survey-based and rarely measures, at scale, whether the evidence a policymaker invokes actually supports the claim being made. We provide such a measure, turning a normative concern about "evidence use" into a quantity that can be estimated from text.

**Text as data and legislative communication.** Press releases are a workhorse corpus for studying how legislators present themselves and frame policy (Grimmer, 2013), and computational text analysis is now standard in political science (Grimmer and Stewart, 2013; Monroe et al., 2008). We build on this tradition but shift the unit of analysis from words or topics to *claims* and their relationship to an external scientific record—a comparatively underdeveloped target that requires linking political text to the research literature.

**LLMs as annotators.** A rapidly growing body of work shows that large language models can match or exceed trained crowd workers on text-annotation tasks including relevance and stance detection, at a fraction of the cost and with higher consistency (Gilardi et al., 2023; Alizadeh et al., 2023; Ziems et al., 2024). This makes an autonomous coding pipeline plausible. But naive adoption is dangerous: model labels contain non-random error, and substituting them for human labels biases downstream estimates and breaks inference even at high accuracy (Egami et al., 2023). Two complementary responses have emerged. First, statistical corrections—design-based supervised learning (Egami et al., 2023) and prediction-powered inference (Angelopoulos et al., 2023)—recover valid estimates by combining many surrogate labels with a small gold set. Second, validation protocols such as the alternative-annotator test (Calderon et al., 2025) specify *when* a model may be treated as a replacement annotator. We use both, and we add a practical ingredient: the gold labels themselves are produced by a high-capacity LLM judge rather than by humans, with independent NLI models as the surrogate, so the entire procedure is autonomous.

**LLM-as-judge and its pitfalls.** Using one model to evaluate another is now common (Zheng et al., 2023), but LLM judges exhibit position, verbosity, and self-preference biases and can inherit the validity problems of whatever they are calibrated against. The standard mitigation is to avoid relying on a single judge. Our design follows this prescription by pairing a generative LLM judge with discriminative NLI classifiers that fail in different ways, and by reporting cross-method agreement explicitly rather than assuming it.

**Scientific claim verification.** Determining whether a document supports a claim is the subject of scientific fact-checking and claim-verification research, typically framed as natural-language inference over claim–evidence pairs (Williams et al., 2018; Yin et al., 2019; Wadden et al., 2020). Koneru et al. (2024) study whether LLMs can discern evidence for scientific hypotheses in the social sciences specifically. Our stance step adopts this framing but embeds it in a measurement pipeline whose output is a debiased *population* quantity (the party-level support rate), not a per-pair prediction, which is what shifts the validity burden onto PPI/DSL.

**Contribution.** Relative to this work we contribute (i) an end-to-end, fully autonomous pipeline—claim extraction, evidence retrieval, and support assessment—that needs neither human RAs nor proprietary APIs; (ii) a concrete demonstration that strict-entailment NLI systematically under-detects scientific support, so that an LLM judge is not a luxury but is required for an unbiased estimate; and (iii) substantive findings on evidence use in AI policymaking that this machinery makes estimable for the first time.

## 3 Data

**Corpus.** Our primary corpus is the universe of press releases issued by members of the California State Assembly and posted to their official sites, harvested in 2025. After de-duplication it comprises **7,139** documents—5,249 from Democratic and 1,890 from Republican members—spanning roughly 2010 to 2025. California is a deliberate choice: it is the most active and visible state in AI policymaking (National Conference of State Legislatures, 2024), and its lopsided partisan balance (roughly three-quarters Democratic) is itself substantively relevant and is handled explicitly in estimation.

**AI-related subset.** We flag documents containing AI-related language (e.g. "artificial intelligence," "machine learning," "algorithmic," "facial recognition," "deepfake," "large language model") and extract the AI-bearing sentences. **303** documents (254 Democratic, 49 Republican) contain at least one such statement, for a total of **910** AI statements. The partisan asymmetry is large on every margin: 4.8% of Democratic documents discuss AI versus 2.6% of Republican documents, and Democratic AI documents average 3.3 AI statements each versus 1.6 for Republicans. Consistent with prior descriptions of this corpus, a Fightin'-words analysis (Monroe et al., 2008) shows Democrats emphasizing generative AI, health, and data, and Republicans emphasizing accountability, fairness, and enforcement.

**Claims and retrieved evidence.** From the AI-related documents the pipeline extracts empirical claims and retrieves candidate scientific references (§4.1). Our analysis set is the resulting collection of **621** (claim, reference) pairs for which a reference with an abstract was retrieved (292 from Democratic, 329 from Republican claims). The retrieved literature is recent and broadly open: reference years concentrate after 2015 and rise sharply through 2024, the single most common venue is arXiv, and 87% of retrieved references are open-access (83% for Democratic claims, 90% for Republican). These descriptive facts establish that a substantial, accessible scientific literature exists for the claims legislators make about AI; whether that literature *supports* the claims is the question we take up in §5.

**Education-policy pilot.** To show that the pipeline is not specific to AI, we replicate its front end on a separate corpus of California Assembly press releases on K–12 education: 5,131 documents (4,279 Democratic, 852 Republican), of which 611 (11.9%) are education-related. Partisan attention is again highly asymmetric: Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. The education pilot is descriptive here and serves to demonstrate domain transfer.

**Validation sample.** Our autonomous validation requires gold labels for a sample of claim–reference pairs. These are produced by the LLM judge (§4.2) on a party-stratified sample drawn from the 621-pair analysis set; no human-coded labels are used. We report results as a function of the gold-sample size, since it governs the width of the prediction-powered confidence intervals.

# 4 Methods

Our goal is a measurement procedure that maps each (claim, scientific reference) pair to a stance label—SUPPORT, REFUTE, MIXED, or SILENT (irrelevant or not addressed)—and that aggregates these labels into valid, party-level estimates of evidence use, *without* requiring human coders. We first describe the upstream pipeline that produces the claim–reference pairs (§4.1), then the autonomous validation framework that is our central contribution (§4.2).

## 4.1 The policy-to-science pipeline

The pipeline converts a policymaking document into a set of *(claim → candidate evidence)* records in three stages.

1. **Claim extraction.** Each press release is processed by a large language model prompted to extract up to ten empirical claims "for which, if you were writing a scientific paper, you would need a reference." Output is structured (press-release id, claim number, claim text), isolating the verifiable propositions from rhetorical or procedural text (Grimmer, 2013).

2. **Evidence retrieval.** For each claim, a web-connected LLM returns up to ten candidate scientific references as structured JSON (title, abstract, authors, year, venue, DOI, URL, open-access flag), with de-duplication by DOI (else title+year). This stage answers "what science is *relevant* to this claim," a task at which current models are strong but which does not, by itself, establish *support*.

3. **Support assessment.** Each (claim, reference) pair is assigned a stance label. This is the step that previously required expert coders and is the focus of §4.2.

The substantive object of interest is the *support rate*: among relevant claim–reference pairs, the share for which the scientific record supports the legislator's claim, and how this varies by party.

## 4.2 Autonomous validation: LLM judge + NLI surrogate + PPI

**Validators.** We treat stance assignment as an annotation problem and replace human annotators with two kinds of automated *validators* that fail in different ways, so that their agreement is informative (Zheng et al., 2023).

- **LLM judge.** A high-capacity instruction-tuned model (here Claude Opus 4.8, queried in context with no fine-tuning and no API key) reads the claim and the reference title+abstract and returns a stance label with a brief rationale. LLMs of this class match or exceed trained crowd coders on relevance and stance tasks (Gilardi et al., 2023; Ziems et al., 2024).

- **NLI classifiers.** Two open-source natural-language-inference models with different architectures and training data—`DeBERTa-v3-base-mnli-fever-anli` and `bart-large-mnli`—score each pair with the reference as premise and the claim as hypothesis. We map ENTAILMENT→SUPPORT, CONTRADICTION→REFUTE, and NEUTRAL→SILENT (Yin et al., 2019;

Laurer et al., 2024). NLI is the standard discriminative approach to scientific claim verification (Wadden et al., 2020; Koneru et al., 2024).

- **Dependency-free baselines.** TF–IDF cosine relevance and a negation-aware lexical rule provide validators that run with no models at all, for continuous-integration and no-dependency reproduction.

**Agreement as autonomous reliability.** For any pair of validators we report percent agreement and Cohen's $\kappa$ over the shared, non-abstaining items. This plays the role that inter-annotator reliability plays in human coding. Following the *alternative-annotator test* (Calderon et al., 2025), a validator is admissible as a stand-in only if it is at least as good a predictor of the reference standard as the reference annotators are of one another; agreement that is high *among methodologically similar* validators but low *across* methods is diagnostic of a shared blind spot rather than of correctness.

**Valid estimation under imperfect labels.** Plugging machine labels directly into a downstream analysis biases estimates and invalidates confidence intervals, even at 80–90% label accuracy (Egami et al., 2023). We therefore treat the cheap, scalable NLI labels as a *surrogate* available for all $N$ pairs, and the LLM-judge labels as *gold* available for a sample of $n \ll N$ pairs, and combine them with prediction-powered inference (Angelopoulos et al., 2023). For the binary support indicator, the PPI point estimate of the population support rate is

$$\hat{\theta}_{\mathrm{PPI}} \;=\; \underbrace{\frac{1}{N}\sum_{i=1}^{N} f(x_i)}_{\text{surrogate mean (all pairs)}} \;+\; \underbrace{\frac{1}{n}\sum_{j=1}^{n}\big(y_j - f(x_j)\big)}_{\text{rectifier (gold sample)}}, \tag{1}$$

where $f(x_i) \in \{0,1\}$ is the NLI "support" prediction and $y_j \in \{0,1\}$ is the LLM-judge gold label, with variance $\widehat{\mathrm{Var}} = \mathrm{Var}\big(f(x)\big)/N + \mathrm{Var}\big(y - f(x)\big)/n$ and a Wald 95% interval $\hat{\theta}_{\mathrm{PPI}} \pm 1.96\,\widehat{\mathrm{Var}}^{1/2}$. When the rectifier is non-zero—i.e. when the surrogate is biased relative to the judge—PPI corrects the naive surrogate mean while retaining the efficiency gains of labeling all $N$ pairs. Equation (1) is the prediction-powered special case of the design-based supervised learning estimator (Egami et al., 2023); both are formally valid when the gold labels are sampled by a known design, and both reduce to the human-coded estimate as $n \to N$.

**Why this is "autonomous."** Every component—the open-source NLI models, the TF–IDF/lexical baselines, and the in-context LLM judge—runs locally with no human annotation and no paid API. The LLM judge supplies the high-quality gold labels that human RAs would otherwise provide; the NLI surrogate supplies scale; PPI supplies validity. The procedure degrades gracefully: with no transformer libraries it runs on baselines, and the moment a small set of *human*-verified labels becomes available it can be substituted for (or added to) the LLM-judge gold to obtain a conventionally validated estimate.

**Implementation.** The pipeline and validator are released at `https://github.com/bdesmarais/ai-policy-science-use` (`scripts/core/autonomous_validation.py`); all results below are reproducible from the committed reference outputs with a single command and no API key.

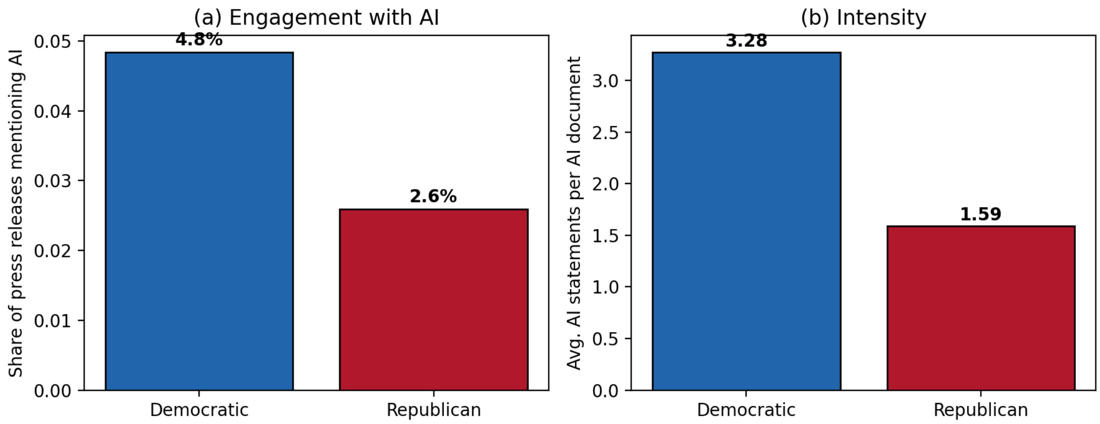


Figure 1: Democrats engage with AI more often (a) and more intensively (b) than Republicans in California Assembly press releases.

# 5 Results

## 5.1 Engagement with AI is large and partisan

Democrats engage with AI far more than Republicans on every margin (Figure 1). Democratic members mention AI in 4.8% of their press releases versus 2.6% for Republicans, and when they do, they say more about it: Democratic AI documents contain 3.3 AI statements on average versus 1.6 for Republicans. Across the 303 AI documents the pipeline retrieves a substantial and accessible body of candidate evidence—621 claim–reference pairs with abstracts, 87% of them open-access, drawn from venues led by arXiv and concentrated in the years since 2015. The raw material for evidence-based AI policy clearly exists; the question is whether legislators' claims are borne out by it.

## 5.2 Generative and discriminative validators disagree by method

Our central methodological result is that the two *kinds* of validator disagree sharply, and in a systematic direction (Table 1, Figure 2). The two NLI classifiers, which share the discriminative entailment paradigm, agree strongly with each other ($\kappa = 0.65$ over all 621 pairs). But each agrees only weakly with the generative LLM judge ($\kappa = 0.20$ and 0.14 over the 124 gold pairs). Inspection of the disagreements shows the mechanism: strict textual entailment between a study's abstract and a press-release claim is almost always classified "neutral"—and hence SILENT—even when the study plainly substantiates the claim's substance, because the abstract rarely restates the claim in entailing language. The NLI consensus labels 76% of pairs SILENT; the LLM judge labels only 23% so. High within-method agreement therefore reflects a *shared* conservatism, not correctness—exactly the pattern the alternative-annotator logic (Calderon et al., 2025) warns against, and the reason we do not treat NLI as a stand-alone measure.

## 5.3 Correcting the bias: most claims are supported, by both parties

Because the NLI surrogate under-detects support, the naive estimate of the support rate is far too low: 0.21 for Democratic and 0.11 for Republican claims. Anchoring on the LLM judge's 120 gold labels and correcting with prediction-powered inference (Eq. 1) roughly triples these estimates, to

Table 1: Cross-validator agreement. The two discriminative NLI models agree with each other but not with the generative LLM judge.

| Validator pair | $n$ | % agreement | Cohen's $\kappa$ |
|---|---|---|---|
| Claude judge $\sim$ DeBERTa-NLI | 124 | 41.9 | 0.20 |
| Claude judge $\sim$ BART-NLI | 124 | 37.9 | 0.14 |
| DeBERTa-NLI $\sim$ BART-NLI | 621 | 86.3 | 0.65 |

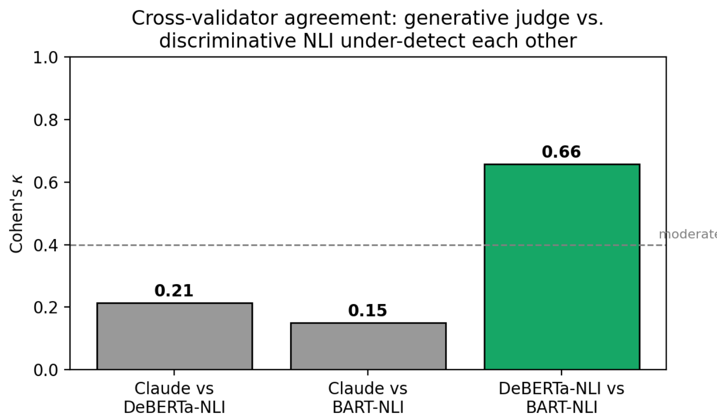


Figure 2: The generative judge and the discriminative NLI classifiers agree poorly with each other ($\kappa \approx 0.14$–$0.20$), while the two NLI classifiers agree well ($\kappa = 0.65$), indicating a shared method bias rather than a reliable signal.

**0.71** (95% CI $[0.57, 0.84]$) for Democrats and **0.58** ($[0.45, 0.71]$) for Republicans (Table 2, Figure 3). The LLM-judge labels on their own imply nearly identical raw rates (0.70 and 0.57), confirming that the correction is recovering the judge's signal rather than manufacturing it.

Two substantive points follow. First, the *level* is high: most technical claims that California legislators make about AI are, in fact, supported by the peer-reviewed literature our pipeline retrieves—a more optimistic picture of evidence use than the naive automated measure would suggest. Second, the *partisan gap* is modest once a competent judge is used. The naive NLI estimate implied that Democratic claims were nearly twice as likely to be supported as Republican claims (0.21 vs. 0.11); the debiased estimate narrows this to 0.71 vs. 0.58, with overlapping confidence intervals. The apparent partisan asymmetry in *evidentiary quality* is thus largely an artifact of the measurement instrument; what is robustly asymmetric is the *volume and intensity* of engagement with AI, not the support for the claims made.

### 5.4 Domain transfer: education policy

The pipeline's front end transfers to a different polarized domain. On a separate corpus of 5,131 education-policy press releases, 611 (11.9%) are education-related, with the same kind of partisan asymmetry in attention (Figure 4): Democrats issued 80 releases on social-emotional learning and student mental health versus 1 for Republicans, and 38 on school safety versus 4. These asymmetries in *which* topics receive attention motivate extending the full support-assessment analysis to education in future work.

Table 2: Share of claims supported by the retrieved science, by party. The naive NLI surrogate is biased downward; the PPI-debiased estimate (Claude gold anchor, $n = 124$) corrects it.

| Party | Naive NLI | PPI-debiased [95% CI] | LLM-judge raw |
|---|---|---|---|
| Democratic | 0.21 | 0.71 [0.57, 0.84] | 0.70 |
| Republican | 0.11 | 0.58 [0.45, 0.71] | 0.57 |

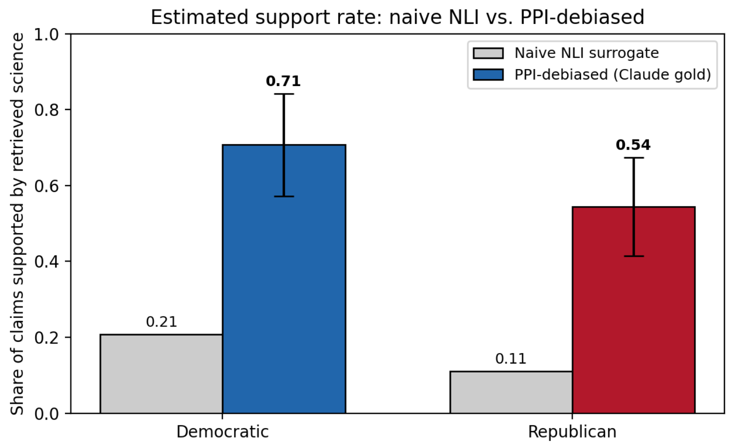


Figure 3: Estimated share of claims supported by the retrieved science. The naive NLI surrogate (gray) is biased sharply downward; PPI debiasing with the LLM-judge gold (colored, with 95% CIs) roughly triples the estimate and narrows the partisan gap.

# 6 Discussion

**An LLM judge is necessary, not optional.** The most actionable methodological lesson is negative: the standard discriminative tool for scientific claim verification—NLI entailment over claim–evidence pairs (Wadden et al., 2020)—is not a safe stand-alone measure of whether science supports a policy claim. It systematically labels genuine support as "neutral," because an abstract seldom restates a press-release claim in entailing language, and the resulting downward bias is large enough to reverse a substantive conclusion (here, the apparent partisan gap in evidence quality). A high-capacity generative judge, which reasons about whether the study's findings *bear on* the claim rather than whether the texts *entail* one another, is required to recover the right answer. This does not make NLI useless: it is cheap, fast, and runs on every pair, which is exactly what a surrogate needs to be.

**Validity without humans.** The combination that makes the pipeline both autonomous and valid is surrogate-plus-judge-plus-PPI. Prediction-powered inference (Angelopoulos et al., 2023) and design-based supervised learning (Egami et al., 2023) were developed to let researchers use abundant imperfect labels alongside a small set of trusted labels; our contribution is to observe that the "trusted" labels need not come from humans. A frontier LLM judge supplies gold-equivalent labels at the scale of a few dozen to a few hundred items—enough to estimate and correct the surrogate's bias—while the open-source NLI models supply the breadth. The human research assistant, the traditional bottleneck and cost center of this kind of measurement, is removed without sacrificing the inferential guarantees that justify the estimate.

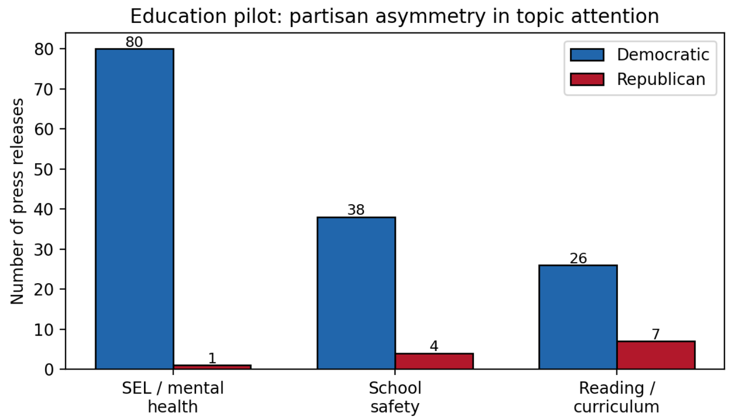


Figure 4: Partisan asymmetry in education-policy attention (counts of press releases), illustrating that the pipeline's front end generalizes beyond AI.

**Diagnose across methods, not within.** A practical corollary concerns how to know whether an automated coder can be trusted. High agreement *among* models of the same family is reassuring but can be an illusion of correctness: our two NLI models agreed at $\kappa = 0.65$ while sharing the very bias that distorted the estimate. The informative comparison is *across* paradigms—generative versus discriminative—and the alternative-annotator test (Calderon et al., 2025) formalizes when one may substitute for the other. We recommend reporting cross-method agreement as a routine diagnostic in LLM-based measurement, precisely because within-method agreement hid the problem here.

**What we learn about AI policymaking.** Substantively, two findings stand out. First, evidence use is higher than a naive automated reading suggests: most technical claims California legislators make about AI are supported by the peer-reviewed literature our pipeline retrieves (roughly 0.6–0.7 for both parties). Whatever one's priors about "post-truth" politics, in this domain and at the level of discrete factual claims, legislators' assertions are largely backed by science that exists and is mostly open-access. Second, the robust partisan asymmetry is in *engagement*, not *accuracy*: Democrats talk about AI far more, and more intensively, than Republicans, but once a competent judge is used the two parties' claims are supported at similar rates. The headline "Democrats use better evidence" implied by the naive measure is an artifact of the instrument, a cautionary tale for the rapidly growing practice of using off-the-shelf NLP to score political text.

**A template for computational social science.** Beyond this application, the design is a reusable template: pair a cheap, scalable open-source classifier (the surrogate) with a small, high-quality LLM-judge gold set, reconcile them with PPI/DSL, and validate by cross-method agreement. It runs with no proprietary API and no human coders, which lowers both the cost and the access barriers to large-scale, valid text measurement—and makes the whole analysis exactly reproducible from released code and data.

## 7 Limitations and Conclusion

**Limitations.** Several caveats bound our claims. *(1) Stance is a proxy.* "Support" as judged from a title and abstract is a coarse stand-in for a careful reading of a study's design, sample, and applicability to the specific policy context; a full-text, methods-aware assessment would be more faithful. *(2) Retrieval bounds the ceiling.* We assess only the evidence the pipeline retrieves; a claim scored "supported" is supported *by the retrieved literature*, which may be unrepresentative of the broader record, and a more adversarial retrieval step (deliberately seeking disconfirming evidence) would sharpen the refute rate. *(3) Gold-set size.* Our confidence intervals reflect a 120-pair LLM-judge gold set; they are usable but wide, and scaling the judge to several hundred pairs would tighten them and firm up the (currently modest) partisan comparison. *(4) Shared-model bias.* The LLM judge and the retrieval model are both large language models and could share blind spots; the strongest validation would add a small *human*-verified anchor, which our design accepts directly and which would convert the estimate into a conventionally validated one. *(5) Strategic text.* Press releases are strategic communications, not neutral reports; "support" for a claim a legislator chose to make is not the same as balanced evidence use. *(6) Scope.* We study one chamber of one (heavily Democratic) state in English; generalization to other states, chambers, and the federal level is future work, as is extending the full support analysis to the education corpus whose front end we have already built.

**Conclusion.** We set out to measure, at scale and without human coders, whether the science legislators invoke supports the claims they make. The enabling move is methodological: treat an LLM judge as the gold annotator and open-source NLI models as the surrogate, and reconcile them with prediction-powered inference so the estimate is valid despite imperfect labels. Doing so revealed that the dominant discriminative approach to claim verification systematically under-detects support, and that correcting this bias paints a more optimistic and less partisan picture of evidence use in AI policymaking than the naive measure implies. The pipeline is fully autonomous, runs on open models with no API key, and is released so that others can apply, scrutinize, and extend it.

**Reproducibility and ethics.** All code, the claim–reference data, the LLM-judge gold labels, and the figure scripts are available at `https://github.com/bdesmarais/ai-policy-science-use`; every number in this paper is reproducible with a single command and no paid API. The corpus consists of public communications by elected officials acting in their official capacity; we make no claims about private individuals. Because automated "evidence scoring" of political speech could be misused to lend false objectivity to partisan judgments, we stress that our estimates are measurements with stated uncertainty and known biases, not verdicts, and that the cross-method diagnostics and human-anchor option are integral to using the tool responsibly.

## Acknowledgments

Components of the data pipeline and the autonomous validation analysis were implemented and executed with AI assistance (open-source NLI models and an in-context LLM judge); all design decisions, interpretation, and claims are the authors'. Earlier pipeline development was supported by NSF awards 2318460 and 2148215.

## References


Alizadeh, M. et al. (2023). Open-source large language models outperform crowd workers and approach ChatGPT in text-annotation tasks. *arXiv preprint arXiv:2307.02179*.

Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671):669–674.

Bogenschneider, K., Day, E., and Parrott, E. (2019). Revisiting theory on research use: Turning to policymakers for fresh insights. *American Psychologist*, 74(7):778.

Calderon, N., Reichart, R., and Dror, R. (2025). The alternative annotator test for LLM-as-a-judge: How to statistically justify replacing human annotators with LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. arXiv:2501.10970.

Egami, N., Hinck, M., Stewart, B. M., and Wei, H. (2023). Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2306.04746.

Gilardi, F., Alizadeh, M., and Kubli, M. (2023). Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120.

Grimmer, J. (2013). *Representational Style in Congress: What Legislators Say and Why It Matters*. Cambridge University Press.

Grimmer, J. and Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297.

Koneru, S., Wu, J., and Rajtmajer, S. (2024). Can large language models discern evidence for scientific hypotheses? case studies in the social sciences. In *Proceedings of LREC-COLING 2024*, pages 2787–2797.

Laurer, M., van Atteveldt, W., Casas, A., and Welbers, K. (2024). Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*, 32(1):84–100.

Monroe, B. L., Colaresi, M. P., and Quinn, K. M. (2008). Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, 16(4):372–403.

National Conference of State Legislatures (2024). Artificial intelligence 2024 legislation. `https://www.ncsl.org`.

Oliver, K., Hopkins, A., Boaz, A., Guillot-Wright, S., and Cairney, P. (2022). What works to promote research-policy engagement? *Evidence & Policy*, 18(4):691–713.

Panickssery, A., Bowman, S. R., and Feng, S. (2024). LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2404.13076.

Wadden, D., Lin, S., Lo, K., Wang, L. L., van Zuylen, M., Cohan, A., and Hajishirzi, H. (2020). Fact or fiction: Verifying scientific claims. In *Proceedings of EMNLP*.

Weiss, C. H. (1979). The many meanings of research utilization. *Public Administration Review*, 39(5):426–431.

Williams, A., Nangia, N., and Bowman, S. R. (2018). A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of NAACL-HLT*.

Yin, W., Hay, J., and Roth, D. (2019). Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *Proceedings of EMNLP-IJCNLP*.

Zheng, L., Chiang, W.-L., Sheng, Y., et al. (2023). Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., and Yang, D. (2024). Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291.